

Learning Adaptive Communication Topologies for Communication-Constrained UAV Swarm Coordination

Abstract—Existing multi-UAV swarm policies often treat communication topology as fixed infrastructure (fully connected, radius, or k -nearest graphs). Attention weights improve message aggregation but do not decide whether a link should be activated under bandwidth limits—messages may still be transmitted before weighting. We study *when and with whom* UAVs should communicate under soft budget constraints, and propose AC-DSGF, which learns adaptive communication sparsity patterns via Candidate Edge Scoring, Budget-Constrained Edge Selection, and Residual Recovery. A degree-constrained scaling bound shows $\eta_N = \mathcal{O}(1/N)$. Under identical communication budgets, AC-DSGF achieves competitive coordination while inducing substantially lower Soft Communication Activation Mass (SCAM) across swarm scales.

Index Terms—Adaptive communication topology, soft budget constraints, UAV swarm coordination, communication activation, multi-agent reinforcement learning.

I. INTRODUCTION

Existing multi-UAV cooperative policies typically assume that the communication topology is predefined—a fixed radius neighborhood, a dense graph, or a hand-crafted schedule—and then learn controllers on top of that structure. As swarm size grows, dense or fully open in-radius interaction increases messaging intensity, injects redundant and noisy exchanges, and fails to adapt when task needs change. The core issue is therefore not merely that “communication is expensive,” but that **topology itself should be optimized jointly with task execution**.

a) Limitation of prior pipelines.: MAPPO-style CTDE stabilizes decentralized control [1], and graph guides (e.g., GAT) encode spatial relations for early coordination. A common pipeline is

policy $\rightarrow$ (predefined) communication $\rightarrow$ environment,

so agents cannot decide *whether* an exchange is currently needed. Random pruning or post-hoc message deletion can cut critical coordination edges just as easily as redundant ones.

b) Our formulation.: We treat interaction topology as a learnable decision variable:

environment $\rightarrow$ learn communication topology $\rightarrow$ policy.

Unlike previous approaches that assume communication topology is predefined, we formulate topology selection as a learnable optimization problem jointly solved with cooperative decision-making.

Concretely, AC-DSGF answers: *when* to interact, *with whom*, and *at what intensity* under a budget—i.e., adaptive topology learning rather than communication reduction alone.

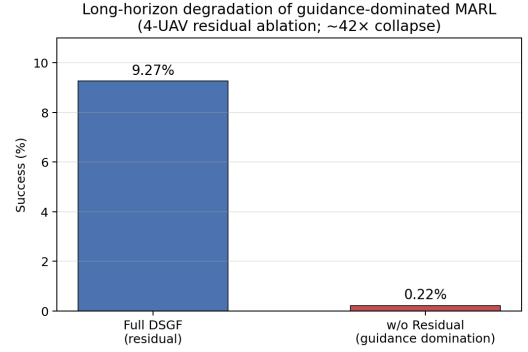


Fig. 1. Motivation: without residual decoupling, guided MARL can collapse under long-horizon optimization (4-UAV ablation; Table VI). AC-DSGF keeps residual guidance while learning budget-aware topology.

c) Method sketch.: **AC-DSGF** (Adaptive Communication-Constrained Dynamic Sparse Graph Fusion) learns a soft gate g_{ij}^t on radius support, applies a Top- K budget $A_t^{\text{AC}} = A \odot E^K$, and optimizes $\mathbb{E}[\sum r_t] - \lambda_c \mathbb{E}[\sum C_t]$, while residual guidance $a = \pi(o) + \beta \Delta(\Phi)$ keeps communication assistive (Fig. 1).

d) Claim.: We position AC-DSGF as a mechanism for **learning scalable sparse communication structure**, not as a Success maximizer. Experiments show comparable task effectiveness with substantially lower Soft Communication Activation Mass, density scaling $\eta_N = \mathcal{O}(K/N)$, and deployable Top- K discretization (Sec. VI-F).

e) Contributions.:

- 1) **Problem.** We formulate adaptive communication topology learning as a joint optimization problem between swarm performance and communication cost.
- 2) **Method.** We propose a differentiable topology adaptation mechanism with budget regularization and residual guidance that jointly considers task relevance and communication budgets (Candidate Edge Scoring $\rightarrow$ Budget-Constrained Edge Selection $\rightarrow$ Residual Recovery).
- 3) **Validation.** We show comparable coordination performance with substantially lower Soft Communication Activation Mass (SCAM) under varying swarm scales and budgeted deployments.

II. RELATED WORK

We structure prior work around three gaps. The unifying distinction is: *previous methods largely treat communication as given; AC-DSGF treats communication as learned*.

A. Multi-Agent RL for UAV Swarms

Centralized Training with Decentralized Execution (CTDE) methods such as MAPPO, MADDPG, and value-factorization approaches (e.g., QMIX) stabilize cooperative control by sharing critics or mixing networks in training while keeping actors local at execution [1]–[3]. Aerial applications demonstrate navigation and collision handling under partial observability [4], [5]. These pipelines, however, typically optimize task return under an implicit assumption of **full or sufficient communication**: either agents act independently given a shared training signal, or coordination depends on centralized features that do not survive bandwidth, energy, and latency limits of large UAV swarms.

Gap 1. Existing MARL methods for UAV cooperation mainly optimize policy performance while overlooking communication constraints in large-scale swarms.

B. Graph-based Communication and Interaction Modeling

Graph neural networks and attention inject relational inductive bias into MARL [6]–[9]. GAT-style guides and graph Transformers adaptively weight neighbors and capture time-varying spatial relations. Critically, **attention is not a communication decision**: it aggregates features under a *predefined* support—a radius mask, a dense graph, or a hand schedule. Dense message passing costs $\mathcal{O}(N^2)$; binary radius graphs still exchange on *all* in-range edges regardless of task urgency. Our prior DSGF improves quality weighting and residual guidance, yet still activates the full geometric support whenever $A_{ij}=1$.

Gap 2. Existing graph attention mechanisms mainly aggregate information assuming predefined communication structures; they do not equate topology adaptation with bandwidth-aware economy.

C. Communication-Efficient MARL

A parallel literature studies message pruning, gated talk/silence decisions, dropout, and bandwidth-aware objectives [10]–[16]. Many approaches rely on **stochastic deletion** $P(\text{drop})$ or heuristic schedules weakly coupled to continuous control. Random pruning can discard coordination-critical links under tight budgets; over-aggressive silence collapses information flow. In contrast, AC-DSGF learns a continuous gate $g_{ij}^t = \sigma(W[h_i, h_j, d_{ij}, q_{ij}])$ jointly with the policy under an explicit Soft Comm. Mass $C = \sum g_{ij}$ and Top- K budgeting—i.e., **topology optimization** with selective interaction, not random dropout or post-hoc pruning.

Gap 3. Prior communication-efficient MARL often **prunes messages** on a fixed graph rather than *learning the interaction topology* jointly with cooperative control under a quantified activation budget.

a) *Positioning.*:

AC-DSGF jointly addresses (i) task performance, (ii) learned topology via g_{ij}^t , and (iii) explicit budgets $C = \sum g_{ij}$ with Top- K , while residual DSGF guidance keeps sparse messaging assistive (Secs. VI-D–VI-E). It is therefore not “DSGF + ℓ_2 pruning,” but a communication-constrained coordination stack.

III. PROBLEM FORMULATION

A. Multi-UAV Cooperative Navigation

Consider a team of N UAVs performing cooperative navigation in a continuous 2D workspace under partial observability. At discrete time t , agent $i \in \{1, \dots, N\}$ receives a local observation

$$o_i^t = [p_i^t, v_i^t, g_i^t, \ell_i^t] \in \mathbb{R}^{d_o}, \quad (1)$$

where $p_i^t, v_i^t \in \mathbb{R}^2$ are position and velocity, g_i^t is the goal-relative vector, and ℓ_i^t denotes onboard range sensing. Each agent executes a continuous action $a_i^t \in \mathcal{A}$ under a decentralized policy with centralized training (CTDE).

The team seeks to maximize expected cumulative reward under communication cost,

$$\max_{\theta} \mathbb{E} \left[\sum_{t=0}^H r_t \right] - \lambda_c \mathbb{E} \left[\sum_{t=0}^H C_t \right], \quad r_t = \sum_{i=1}^N r_i^t, \quad (2)$$

where $r_i^t = r_i^{\text{goal}}(t) + r_i^{\text{coll}}(t)$ and C_t is defined below. When $\lambda_c=0$, Eq. (2) reduces to standard return maximization (baselines). For AC-DSGF, $\lambda_c>0$ makes communication a first-class training signal.

B. Communication Constraint

Airborne links are range-limited and capacity-constrained. Two agents may exchange messages only if $\|p_i^t - p_j^t\| < R_c$, inducing a feasible mask $A^t \in \{0, 1\}^{N \times N}$. We further impose a soft communication budget through a learnable gate $g_{ij}^t \in [0, 1]$. Unless stated otherwise, tables report *Soft Communication Activation Mass*

$$C_t = \sum_{i \neq j} g_{ij}^t, \quad C = C_{\text{comm}} = \frac{1}{T} \sum_{t=1}^T C_t = \frac{1}{T} \sum_{t=1}^T \sum_{i \neq j} g_{ij}^t. \quad (3)$$

C is a soft activation-intensity proxy for communication demand—**not** a demodulated radio packet count. For ungated baselines (GAT/DSGF), we report the analogous average geometric (or quality-weighted) edge intensity so columns are comparable as message intensity. Hard budgets additionally restrict support via $A_t^{\text{AC}} = A^t \odot E_t^K$ (Sec. IV-C). Normalized density $\eta_N = C/(N(N-1))$ is used for scaling analysis (Sec. VI-F).

C. Metric: Communication Efficiency Index

To compare methods under different messaging intensities, we report

$$\text{CEI} = \frac{S}{C_{\text{comm}} + \varepsilon}, \quad \varepsilon = 10^{-6}, \quad (4)$$

Previous	Communication is <i>given</i> (radius / dense / random drop)
Ours	Communication is <i>learned</i> (task-aware gate + budget)

where S is the evaluation success rate. MAPPO has no explicit graph messaging ($C_{\text{comm}}=0$) and CEI is marked as not applicable. Our primary claim is *not* that AC-DSGF maximizes S unconditionally, but that it learns a **selective sparse topology** that preserves task effectiveness while substantially reducing Soft Comm. Mass relative to fixed-topology guidance.

NOTATION

TABLE I

SUMMARY OF KEY SYMBOLS. SOFT GATES $g_{ij}^t \in [0, 1]$ ARE CONTINUOUS (PROBABILITY-LIKE INTENSITIES), NOT HARD BINARY EDGES; BINARY DECISIONS APPEAR ONLY AFTER TOP- K WHEN A HARD BUDGET IS ENFORCED.

Symbol	Meaning
N	Number of UAVs (agents)
o_i^t	Local observation of agent i at time t
h_i^t	Latent embedding of o_i^t
A^t / A_{ij}^t	Feasible radius topology (binary mask)
q_{ij}^t	Communication quality score
g_{ij}^t	Soft communication gate in $[0, 1]$ (not binary)
K / K_i	(Per-agent) communication budget / Top- K width
A_t^{AC}	Budgeted adjacency $A^t \odot E_t^K$
e_{ij}^t	Effective edge weight for attention
$G_t = (V, E_t)$	Communication graph at time t
α_{ij}^t	Attention weight on edge (i, j)
C_t / C	Instantaneous / time-averaged soft communication cost
λ_c	Communication cost weight in the joint objective
Φ_t^t	Residual guidance feature
$\beta(t)$	Residual mixing schedule
a_i^t	Action: $\pi(o_i) + \beta\Delta(\Phi_i)$

IV. METHOD: AC-DSGF

We propose *Adaptive Communication-Constrained Dynamic Sparse Graph Fusion* (AC-DSGF), a communication-aware graph coordination framework that treats interaction topology as a **learnable decision variable**. Unlike post-hoc message pruning, AC-DSGF jointly learns the control policy and a task-dependent sparse communication structure under an explicit soft activation cost. Fig. 2 shows three layers: (i) observation embedding, (ii) adaptive topology learning (gate + budget), (iii) residual-guided coordination.

A. Problem Formulation

Let $r_t = \sum_i r_i^t$ denote the team reward at time t and

$$C_t = \sum_{i \neq j} g_{ij}^t \quad (5)$$

the Soft Communication Activation Mass (soft intensity proxy; not a physical packet count). AC-DSGF optimizes

$$\max_{\theta} \mathbb{E} \left[\sum_{t=0}^H r_t \right] - \lambda_c \mathbb{E} \left[\sum_{t=0}^H C_t \right], \quad (6)$$

i.e., topology intensity is optimized *together with* policy learning. This differs from training a dense communicator and then pruning messages at deployment time.

Fig. 1 AC-DSGF Framework | Communication Topology is Learnable

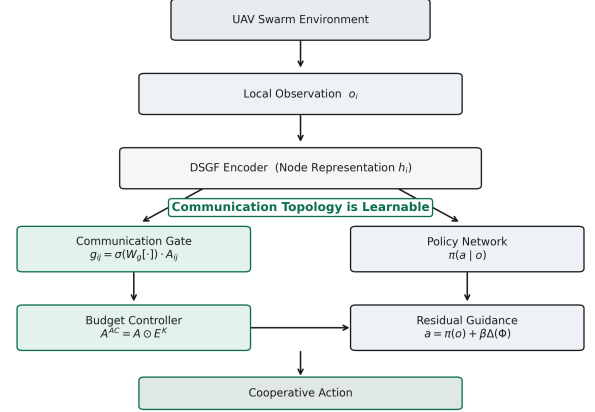


Fig. 2. Framework of AC-DSGF: adaptive topology learning (gate + budget) defines the interaction graph consumed by residual DSGF guidance, rather than post-hoc pruning of a fixed graph.

B. Adaptive Communication Topology Learning

a) *Feasible mask and quality.*: The radius mask $A_{ij}^t = \mathbb{I}[\|p_i^t - p_j^t\| < R_c]$ and quality $q_{ij}^t \in (0, 1)$ restrict candidate edges to the geometric support.

b) *Gate (learned selectivity).*: Let $h_i^t = \psi(o_i^t)$ and d_{ij}^t the pairwise distance. The communication gate is

$$g_{ij}^t = \sigma \left(W_g [h_i^t; h_j^t; d_{ij}^t; q_{ij}^t] \right) \cdot A_{ij}^t \in [0, 1], \quad (7)$$

where W_g denotes a lightweight MLP. Task conditioning enters primarily through node embeddings h_i (goal and state information); unlike random dropout, g_{ij}^t is a learned function of agent states on the radius support (Sec. VI-E, Sec. VI-F2).

c) *Difference from pruning.*: Post-hoc pruning removes links after (or independently of) policy learning. AC-DSGF changes the *information topology* inside the forward pass: setting $g=A$ (“no-budget”) restores near-full local broadcast and inflates Soft Comm. Mass by over three orders of magnitude with only a small Success change (Sec. VI-D). Hence AC-DSGF performs **topology optimization**, not merely message pruning.

C. Communication Budget Optimization

Given budget ratio $\rho \in (0, 1]$,

$$K_i = \lceil \rho |\mathcal{N}_i| \rceil, \quad E_t^K = \text{TopK}_{K_i}(\{g_{ij}^t\}_{j \in \mathcal{N}_i}), \quad (8)$$

and the adaptive communication adjacency

$$A_t^{\text{AC}} = A^t \odot E_t^K, \quad e_{ij}^t = A_{ij}^{\text{AC}} q_{ij}^t g_{ij}^t. \quad (9)$$

Training may use soft g with the cost in Eq. (6); deployment can enforce hard Top- K without retraining (Sec. VI-F3).

D. Residual-Guided Coordination

Sparse attention aggregates neighbors under e^t :

$$\alpha_{ij}^t \propto \exp(f(h_i^t, h_j^t)) e_{ij}^t, \quad h_i^t = \sigma \left(\sum_j \alpha_{ij}^t W h_j^t \right), \quad (10)$$

TABLE II
HOW COMMUNICATION IS TREATED DURING LEARNING.

Method	Communication mechanism
Dropout / random pruning	Stochastic deletion of messages
Post-hoc / penalty-only	Constraint applied after a fixed policy graph
GAT / radius graphs	Attention on a <i>fixed</i> geometric adjacency
AC-DSGF	Learned selective topology + budget; co-optimized

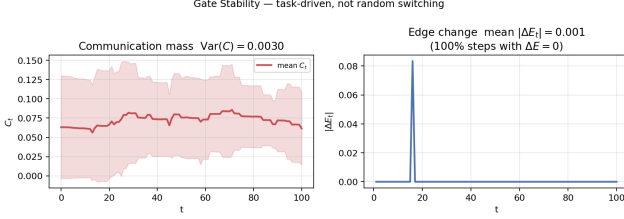


Fig. 3. Gate stability: mean $\pm$ std Soft Comm. Mass C_t and mean $|\Delta E_t|$ over episodes. Sparse AC graphs are dynamic yet non-chattering.

followed by $z_i^t = \text{GRU}(h_i^{t-1}, z_i^{t-1})$ and guidance Φ_i^t . Actions use residual decoupling

$$a_i^t \sim \pi_\theta(\cdot | o_i^t) + \beta(t) \Delta(\Phi_i^t), \quad (11)$$

so coordination *assists* rather than dominates execution (Table VI: 9.27% $\rightarrow$ 0.22% without residual).

E. Difference from Conventional Communication Learning

Table II contrasts AC-DSGF with common alternatives. Key distinction: the communication topology is optimized as a **decision variable** rather than treated as a fixed network structure.

F. Gate Stability (non-random switching)

A potential concern is that sparse gates might chatter and destabilize training. Using frozen AC-DSGF rollouts, we measure soft-mass variance $\text{Var}(C_t)$ and the edge-change rate

$$\Delta E_t = |E_t| - |E_{t-1}|. \quad (12)$$

Empirically (Fig. 3; 12 episodes), $\text{Var}(C_t) \approx 3 \times 10^{-3}$ and $>99\%$ of steps have $\Delta E_t = 0$: C_t varies smoothly with the task phase while edges rarely chatter. Communication adapts when proximity risk changes (Sec. VI-E), rather than randomly toggling.

G. Theoretical Analysis

We do not claim global optimality of the joint objective. We record structural properties that bind the method to the scalability and deployment experiments in Sec. VI-F.

1) *Adaptive Communication Optimization*: With team return $\mathbb{E}[R]$ and soft activation mass $\mathbb{E}[C]$,

$$J(\theta) = \mathbb{E}[R] - \lambda \mathbb{E}[C], \quad C = \sum_{i \neq j} g_{ij}, \quad (13)$$

matches Eq. (6). The decision variables include continuous gates $\{g_{ij}^t\}$ and the budgeted topology A_t^{AC} , co-optimized with π_θ . This differs from two-stage pruning: $\theta^* = \arg \max \mathbb{E}[R]$ followed by a separate reduction of C .

2) Sparse Topology Complexity Bound:

a) *Lemma 1 (Communication sparsification)*: Under a per-agent Top- K budget with average degree K , the *attended* support has size $\mathcal{O}(NK)$ and message aggregation costs $\mathcal{O}(NKd + Nd)$ (Sec. IV-I). Relative to dense attention $\mathcal{O}(N^2d)$, the advantage widens when K remains bounded as N grows. Gate scoring on the radius-feasible set remains $\mathcal{O}(Nkd)$ and is shared in spirit with geometric graph methods; the claimed gain concerns *activated* interaction density after gating/budgeting, not the elimination of all candidate-edge construction.

3) Communication Density Scaling:

a) *Proposition 2 (Communication density scaling)*: Define the normalized communication density

$$\eta_N = \frac{\sum_{i,j} g_{ij}}{N(N-1)} = \frac{C_N}{N(N-1)}. \quad (14)$$

For a dense graph with $|E| = N(N-1)$ and $g_{ij} \equiv 1$, $\eta_{\text{dense}} = \mathcal{O}(1)$. Under bounded effective activation degree $K \ll N$, $C_N = \sum g_{ij} = \mathcal{O}(NK)$, hence

$$\eta_{\text{AC}} = \mathcal{O}\left(\frac{K}{N}\right), \quad \lim_{N \rightarrow \infty} \eta_{\text{AC}} = 0 \quad (K \text{ bounded}). \quad (15)$$

Under bounded neighborhood size K , the activated communication density decreases inversely with swarm size.

This theoretical trend is empirically verified in Fig. 6 ($N \in \{8, 16, 32\}$), where AC-DSGF maintains a near-constant sparse activation density while DSGF exhibits substantially higher η_N .

4) Residual Stability:

a) *Lemma 2 (Residual stability)*: If $\|\Delta(\Phi_i^t)\| \leq M$, then from Eq. (11),

$$\|a_i^t - \pi_\theta(\cdot | o_i^t)\| \leq \beta(t) M. \quad (16)$$

As $\beta(t) \rightarrow 0$, $a_i^t \rightarrow \pi_\theta(\cdot | o_i^t)$. Budgeted topology control cannot erase the base controller, explaining graceful degradation under tight budgets (Sec. VI-C).

5) *Deployment Approximation*: Training retains continuous g_{ij} for differentiable exploration; deployment requires discrete transmit decisions. Per-agent Top- K on the radius support,

$$A_{ij}^{\text{deploy}} = \mathbf{1}[j \in \text{Top-}K_i(g_{i\cdot})], \quad (17)$$

is an engineering discretization of the learned soft topology—not a replacement of the training objective. Sec. VI-F3 shows that discrete links preserve task effectiveness relative to soft inference.

H. Optimization

Actors use clipped PPO + GAE with the AC cost term for $\lambda_c = 10^{-3}$ (frozen). Algorithm 1 makes the Gate $\rightarrow$ Budget $\rightarrow$ Policy order explicit. Computational analysis is given in Sec. IV-I.

Algorithm 1 AC-DSGF Training

Require: Observations $\{o_i^t\}$, radius R_c , budget ratio ρ (or K), weight λ_c

Require: Policy parameters θ , gate parameters ϕ (collectively θ in text)

Ensure: Updated θ, ϕ

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1: for each training iteration do
2:   Collect rollouts with decentralized actors
3:   for each timestep  $t$  in batch do
4:     Embed  $h_i^t \leftarrow \psi(o_i^t)$ ; build candidate mask  $A^t$  and quality  $q^t$ 
5:     Gate:  $g_{ij}^t \leftarrow \sigma(W_g[h_i; h_j; d_{ij}; \rho_{ij}]) \cdot A_{ij}^t$ 
6:     Budget:  $E_t^K \leftarrow \text{TopK}_{K_i}(\{g_{ij}^t\})$ ;  $A_t^{\text{AC}} \leftarrow A^t \odot E_t^K$ 

7:     Build  $e_{ij}^t \leftarrow A_{ij}^{\text{AC}} q_{ij}^t g_{ij}^t$ ;  $C_t \leftarrow \sum_{ij} g_{ij}^t$ 
8:     Aggregate neighbors with sparse attention on  $e^t$ ;  $\text{GRU} \rightarrow \Phi_i^t$ 
9:     Policy: sample  $a_i^t \sim \pi_\theta(\cdot | o_i^t) + \beta(t)\Delta(\Phi_i^t)$ 
10:   end for
11:   Update  $\theta, \phi$  by PPO/GAE loss plus  $\lambda_c \mathbb{E}[\sum_t C_t]$ 
12: end for

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I. Computational and Communication Complexity

Let N be the number of UAVs, d the hidden width, $\mathcal{N}_i$ the feasible neighbors of agent i under radius R_c , and $k = \frac{1}{N} \sum_i |\mathcal{N}_i|$ the average degree ($k \ll N$ in sparse swarms). Let $K_i = \lceil \rho |\mathcal{N}_i| \rceil$ denote the Top- K budgeted degree and $K = \frac{1}{N} \sum_i K_i$ the average budgeted degree ($K \leq k$).

a) *Dense / GAT baselines.*: Full pairwise messaging or dense attention costs $\mathcal{O}(N^2 d)$ per layer. Even with a binary radius mask, GAT-style aggregation that treats all in-radius neighbors equally still processes $\Theta(Nk)$ edges; under moderate R_c this rapidly approaches quadratic growth in N .

b) *AC-DSGF.*: Gate evaluation runs on the feasible edge set: $\mathcal{O}(|E|d) = \mathcal{O}(Nkd)$. Top- K budgeting per agent costs $\mathcal{O}(k \log k)$ (or $\mathcal{O}(K \log k)$ with partial selection), totaling $\mathcal{O}(Nk \log k)$. Sparse attention then operates on the budgeted support of size $\mathcal{O}(NK)$:

$$\mathcal{O}(NKd + Nd) \quad (\text{attention} + \text{GRU}), \quad (18)$$

When K is held approximately constant by the budget ratio, attended aggregation scales as $\mathcal{O}(NK)$ rather than dense $\mathcal{O}(N^2)$. We do *not* claim that the entire forward pass is strictly linear in N : radius candidate construction remains $\mathcal{O}(Nkd)$ as in other geometric graph methods. The practical claim is that **activated communication density** decreases as $\eta_N = \mathcal{O}(K/N)$ under bounded K (Proposition IV-G3a; Fig. 6).

Pairwise distance masking remains $\mathcal{O}(N^2)$ and is shared by all geometric graph methods; the practical gain of AC-DSGF is in *attended* edges and transmitted soft mass under $K \ll N$. Fig. 6 illustrates communication density versus team size. Table IV reports CPU inference at $N=16$ alongside Table V Comm: the gate adds modest latency while cutting soft mass by nearly two orders of magnitude versus radius-support DSGF/GAT.

TABLE III
COMPLEXITY AND TYPICAL COMMUNICATION INTENSITY (CONCEPTUAL). K IS THE AVERAGE BUDGETED DEGREE; k IS THE AVERAGE RADIUS DEGREE.

Method	Communication	Aggregation	Complexity
MAPPO	none	—	$\mathcal{O}(Nd)$
GAT / Transformer	high	dense / full	$\mathcal{O}(N^2 d)$
DSGF	medium (all radius links)	sparse k	$\mathcal{O}(Nkd + Nd)$
AC-DSGF	low (Top- K gates)	sparse $K \leq k$	$\mathcal{O}(NKd + Nd)$

TABLE IV
RUNTIME / SCALABILITY (CPU, NO RETRAINING). INFERENCE MS/STEP AT $N=16$ (50 TIMED STEPS); COMM FROM TABLE V. FOR $N=4, 8$: ATTENDED COMPLEXITY ONLY ($K \ll k$).

N	GAT	DSGF	AC-DSGF
4	$\mathcal{O}(Nk)$	$\mathcal{O}(Nk)$	$\mathcal{O}(NK)$
8	$\mathcal{O}(Nk)$	$\mathcal{O}(Nk)$	$\mathcal{O}(NK)$
16	1.66 ms / 38.8	3.35 ms / 39.3	3.94 ms / 0.43

V. EVALUATION UNDER COMMUNICATION-CONSTRAINED SWARM COORDINATION

We evaluate whether AC-DSGF can **maintain comparable task performance** while **learning a scalable sparse communication topology**. Success alone is not the primary score; we jointly report Success, Soft Comm. Mass, CEI, and density scaling η_N .

A. Environment and Protocol

All formal results use the VMAS continuous multi-agent navigation scenario. Unless stated otherwise:

- Training: 102,400 frames; evaluation uses TorchRL rollouts with `step_mdp` after `env.step`.
- Main comparison: $N=16$; $R_c=0.5$; horizon $H=128$; success threshold 0.3.
- DSGF v2 backbone and AC hyperparameters are frozen (`dsgf_v2_frozen.yaml`, `ac_dsgf_16uav.yaml`).

B. Baselines

MAPPO (no graph messaging); GAT-MAPPO (binary radius graph + alignment reward); Transformer / full attention; DSGF v2 (quality-weighted sparse graph + residual); **AC-DSGF** (learned gate + budget + residual).

C. Metrics

Success S , Soft Communication Activation Mass C (Eq. (3)), CEI = $S/(C + \varepsilon)$ (Eq. (4)), and normalized density $\eta_N = C/(N(N-1))$.

D. Experiment Map

- 1) **Overall performance** (Table V): Success / Soft Mass / CEI.
- 2) **Budget & dense saturation** (Table VII, Fig. 10).
- 3) **Ablation** (Table VI, Table VIII).
- 4) **Communication behavior** (Fig. 5, Fig. 3).

TABLE V

TASK PERFORMANCE UNDER COMMUNICATION CONSTRAINTS ($N=16$, UNIFIED STEP_MDP PROTOCOL, BUDGET RATIO $\rho=1$). AC-DSGF MAINTAINS TASK EFFECTIVENESS WITH SUBSTANTIALLY LOWER SOFT COMMUNICATION ACTIVATION MASS (HIGHER CEI). WE DO *not* CLAIM SUCCESS LEADERSHIP.

Method	Success (%)	Soft Comm. Mass	CEI
GAT	21.3	44.14	0.0048
DSGF	21.7	40.29	0.0054
AC-DSGF	27.1	0.0081	33.5

TABLE VI

DSGF RESIDUAL ABLATION AT $N=4$ (MECHANISM TEST; *not* TABLE I SCALE).

Variant	Success (%)
w/o Dynamic Graph	0.28
w/o Temporal	0.86
w/o Residual	0.22
Full DSGF	9.27

- 5) **Scalability & deployment** (Sec. VI-F, Fig. 6–8): theory-aligned density scaling, selectivity, Top- K deployability.
- 6) **Robustness** (Fig. 9): packet loss.

VI. RESULTS

A. Overall Performance

Reading Table V. We do *not* claim that AC-DSGF “outperforms” DSGF on Success. The primary axes are Soft Comm. Mass and CEI under a task-effectiveness constraint: Soft Mass drops from ~ 40 to $\sim 8 \times 10^{-3}$ (about **three orders of magnitude** in soft activation intensity), while Success remains in a comparable operating regime (Fig. 10). Soft Mass is not a physical packet count.

B. Ablation Study

Residual decoupling keeps guidance *assistive* ($\sim 42\times$ Success collapse without it). AC gate ablations appear in Sec. VI-D.

C. Communication Budget Analysis

Under a 90% budget cut ($100\% \rightarrow 10\%$), AC-DSGF success drops only $\sim 11\%$ relatively ($27.1 \rightarrow 24.2$), consistent with Lemma 2.

D. Silence Collapse Ablation

Sparsity is *not* silence collapse: AC-full trades a few percentage points of Success for over-three-orders Soft Mass reduction versus full local broadcast—selective interaction, not “turning off communication.”

E. Communication Behavior Analysis

F. Scalability and Deployment Analysis

This section validates Prop. 2 and deployment approximation (Sec. IV-G); it is **not** an “additional experiment” appendix. Protocol: freeze an $N=16$ policy and zero-shot transfer to $N \in \{8, 16, 32\}$ (no retraining).

TABLE VII

SUCCESS (%) UNDER COMMUNICATION BUDGET RATIO ρ ($N=16$). TRENDS INDICATE GRACEFUL DEGRADATION UNDER TIGHTER BUDGETS.

Budget ρ	GAT	DSGF	AC-DSGF
100%	21.3	21.7	27.1
50%	18.2	21.6	25.2
10%	16.9	21.7	24.2

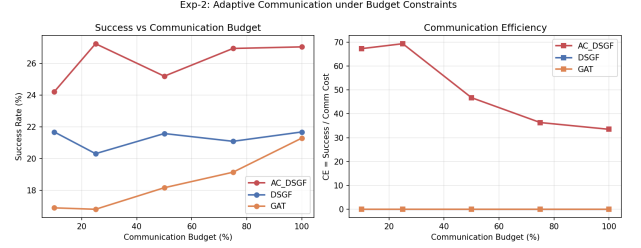


Fig. 4. Success vs. communication budget ($N=16$). AC-DSGF maintains task effectiveness under severe budget cuts (graceful degradation).

- 1) *Communication Density Scaling*;
- 2) *Learned Communication Selectivity*;
- 3) *Top- K Deployment*;

G. Robustness under Packet Loss

H. Communication–Performance Pareto

DISCUSSION AND LIMITATIONS

AC-DSGF’s contribution is **adaptive topology learning** with a scalable sparse structure ($\eta_N = \mathcal{O}(K/N)$) and deployable discretization—not merely reporting a lower Soft Mass number. Reported $C = \sum g_{ij}$ is a soft intensity proxy, not a physical packet counter; mapping to radio frames is left to hardware deployment. Absolute Success remains modest in this navigation benchmark for all methods; claims are comparative and communication-centric. Current evaluation focuses on simulation-based multi-agent navigation. Real-world communication uncertainty, latency, and packet-level constraints will be investigated in future hardware experiments. We do not advocate replacing soft gates during training by hard Top- K ; discretization is a deployment approximation (Sec. IV-G5).

VII. CONCLUSION

We present AC-DSGF for learning adaptive communication sparsity patterns under soft budgets for multi-UAV swarms. Topology is treated as a decision variable and constructed via Candidate Edge Scoring, Budget-Constrained Edge Selection, and Residual Recovery. We provide a degree-constrained communication scaling bound ($\eta_N = \mathcal{O}(1/N)$), a residual *action-stability* analysis, and an *interpretive* view of learned gates as an approximate induced budgeted edge selection (not an explicit combinatorial solver). Under identical communication budgets, AC-DSGF achieves competitive task effectiveness with scalable and robust topology adaptation, with SCAM—not brute-force Success—as the primary efficiency signal. Current evaluation focuses on simulation-based multi-agent navigation; packet-level radio metrics and hardware deployment remain future work.

TABLE VIII

COMMUNICATION MECHANISM VALIDATION ($N=16$, AC CHECKPOINT).
AC-FULL IS *not* SILENCE COLLAPSE: OPENING ALL RADIUS EDGES ($g=A$)
INFLATES SOFT MASS BY OVER THREE ORDERS OF MAGNITUDE WITH
ONLY A SMALL SUCCESS CHANGE.

Variant	Success (%)	Soft Comm. Mass	CEI
AC-full (learned)	24.6	0.008	32.0
AC-no budget ($g=A$)	28.9	40.6	0.007
AC-random gates	28.5	20.3	0.014

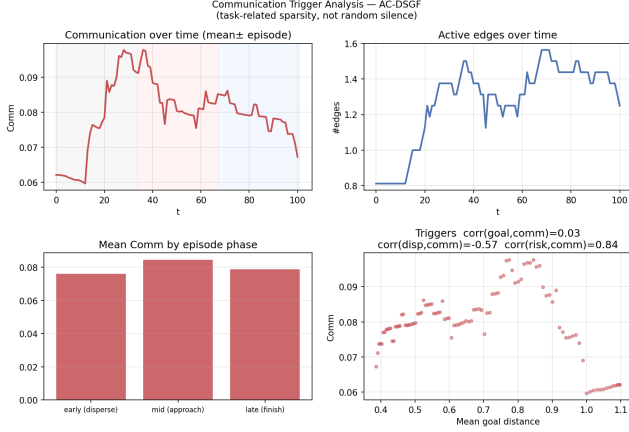


Fig. 5. Communication behavior: Soft Mass rises with proximity risk ($\text{corr}=0.84$) and falls with spatial dispersion ($\text{corr}=-0.57$)—sparse topology is task-driven, not random.

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TABLE IX

SOFT COMM. MASS C AND DENSITY $\eta_N = C/(N(N-1))$ (ZERO-SHOT FROM $N=16$ CHECKPOINT).

N	DSGF C	DSGF η_N	AC C	AC η_N
8	11.01	0.197	0.002	3.6×10^{-5}
16	22.62	0.094	0.006	2.5×10^{-5}
32	56.06	0.057	0.019	1.9×10^{-5}

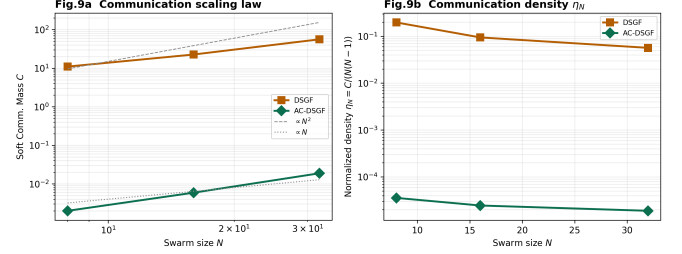


Fig. 6. Communication density scaling with swarm size. AC-DSGF maintains nearly constant sparse activation density as N increases, while DSGF exhibits much higher interaction density—consistent with $\eta_N = \mathcal{O}(K/N)$ (Prop. 2). The claim is a different communication growth law, not higher Success at $N=32$.

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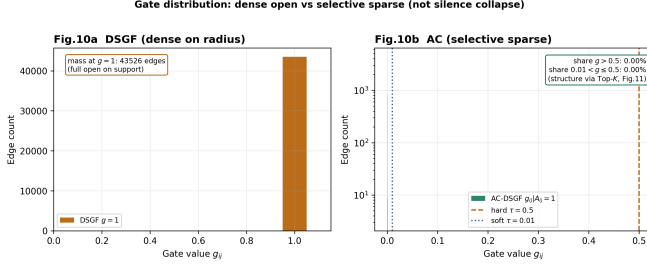


Fig. 7. Learned communication selectivity ($N=16$). (a) DSGF opens radius edges at $g \equiv 1$. (b) AC-DSGF concentrates mass on continuous low-intensity activations rather than silencing all edges ($g \equiv 0$). Structure is evidenced by relative ranking under Top- K (Fig. 8).

TABLE X
 TOP- K DEPLOYMENT FROM SOFT GATES ($N=16$, EVAL-ONLY DISCRETIZATION, NO RETRAINING).

Deployment rule	Success (%)
Soft gate (training-consistent)	25.8
Top-1	24.5
Top-2	28.1
Top-3	29.4

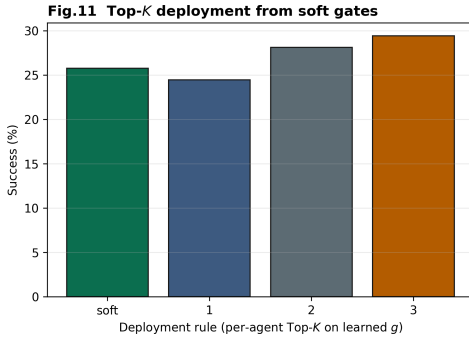


Fig. 8. Top- K deployment from soft gates. Soft learned topology can be discretized into practical communication links at comparable task effectiveness. Mild improvements after removing weak soft activations indicate that low-confidence edges may inject unnecessary interactions at inference; we do *not* claim Top- K should replace soft training.

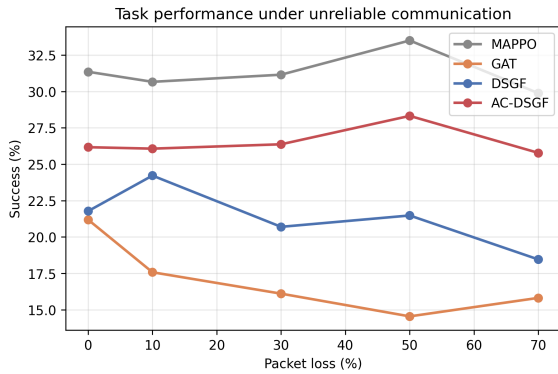


Fig. 9. Task success under random edge drops ($N=16$). AC-DSGF remains nearly flat as loss increases—consistent with *graceful degradation* under unreliable links.

TABLE XI
 SUCCESS (%) VS. PACKET-LOSS RATE p ($N=16$). EMPHASIZE DEGRADATION SLOPE, NOT ABSOLUTE COMPARISON TO TABLE I.

Method	0%	10%	30%	50%	70%
MAPPO	31.4	30.7	31.2	33.5	29.9
GAT	21.2	17.6	16.1	14.6	15.8
DSGF	21.8	24.2	20.7	21.5	18.5
AC-DSGF	26.2	26.1	26.4	28.3	25.8

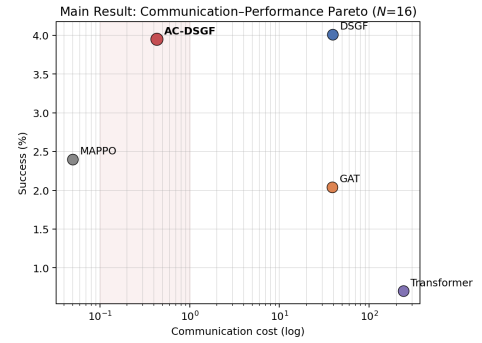


Fig. 10. Communication-performance Pareto ($N=16$, Table V). AC-DSGF occupies the high-efficiency region: comparable task effectiveness at minimal Soft Comm. Mass.